

PRACTICAL-2

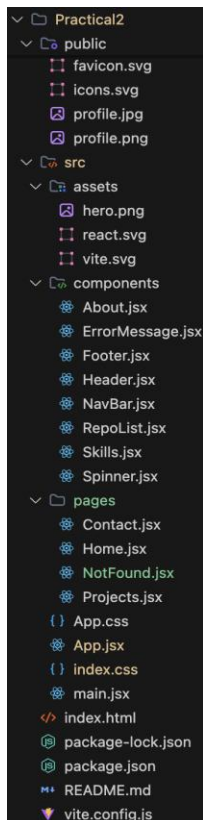
PROBLEM DEFINITION:

- Extend the portfolio application from Practical 1 by adding React Router.
- Implement a navigation bar with at least 3 routes: Home, Projects, and Contact.
- Each route must render a distinct component without a full page reload.
- Add at least 2 useState variables used meaningfully one for toggling UI visibility and one for managing a form input on the Contact page.
- The Contact page must include a controlled input that captures and displays user input in real time.

SUPPLEMENTARY PROBLEMS:

- Add a 404 Not Found route that renders a custom error component for unknown paths.
- Implement a dark/light mode toggle using useState that applies a CSS class to the root element.
- Store the contact form input in state and display a live character count below the input.

ARCHITECTURE / DIAGRAM:



OUTPUT (SUPPLEMENTARY PROBLEMS INCLUDED):

 **Jainam Kamani**

[Home](#) [Projects](#) [Contact](#) 

Open for IT & Software Engineering Internships

 Student Portfolio

**Hello, I'm Jainam Kamani**

Information Technology Student & Web Developer

 Charotar University of Science and Technology (CHARUSAT)  B.Tech Information Technology

 **About Me**

I am an Information Technology student at Charotar University of Science and Technology, originally from Junagadh, Gujarat. I am passionate about building modern, interactive web applications using technologies like React, JavaScript, and Vite, and constantly expanding my software development skills.

7.7 / 10
Cumulative CGPA

2
Projects Built

CHARUSAT
University

Junagadh
Hometown (Gujarat)

 **Academic Coursework & Honors**

Key Information Technology courses and academic achievements at CHARUSAT.

IT301 Advanced Web Frameworks- Grade A+

IT305 Data Structures & Algorithms- Grade A

IT410 Software Engineering Principles- Grade A

IT425 Database Management Systems- Grade A-

 **Skills & Expertise**

Frontend Development

React.jsJavaScript (ES6+)HTML5 & CSS3ViteTailwind CSSBootstrap

Backend & Databases

Node.jsExpress.jsRESTful APIsSQLMongoDB

Tools & Workflows

Git & GitHubVS CodeNpm/VitePostman

Core Competencies

Web Development FrameworksData StructuresOOPLResponsive Web Design

 **Jainam Kamani**

[Home](#) [Projects](#) [Contact](#) 

Open for IT & Software Engineering Internships

 Student Portfolio

**Hello, I'm Jainam Kamani**

Information Technology Student & Web Developer

 Charotar University of Science and Technology (CHARUSAT)  B.Tech Information Technology

 **About Me**

I am an Information Technology student at Charotar University of Science and Technology, originally from Junagadh, Gujarat. I am passionate about building modern, interactive web applications using technologies like React, JavaScript, and Vite, and constantly expanding my software development skills.

7.7 / 10
Cumulative CGPA

2
Projects Built

CHARUSAT
University

Junagadh
Hometown (Gujarat)

 **Academic Coursework & Honors**

Key Information Technology courses and academic achievements at CHARUSAT.

IT301 Advanced Web Frameworks- Grade A+

IT305 Data Structures & Algorithms- Grade A

IT410 Software Engineering Principles- Grade A

IT425 Database Management Systems- Grade A-

 **Skills & Expertise**

Frontend Development

React.jsJavaScript (ES6+)HTML5 & CSS3ViteTailwind CSSBootstrap

Backend & Databases

Node.jsExpress.jsRESTful APIsSQLMongoDB

Tools & Workflows

Git & GitHubVS CodeNpm/VitePostman

Core Competencies

Web Development FrameworksData StructuresOOPLResponsive Web Design

Let's Connect

Feel free to reach out for collaborations or project inquiries.

 jainamkamani95@gmail.com  [GitHub](#)  [LinkedIn](#)

© 2026 Jainam Kamani. All rights reserved.
 Junagadh, Gujarat, India

 Jainam Kamani

Home Projects Contact

My Featured Projects

Explore my completed software projects in AI Healthcare and Financial Analytics.

All AI & Healthcare FinTech & Analytics

AI & Healthcare★ Featured Project

AI Healthcare Management System
An intelligent healthcare platform leveraging AI for automated patient diagnostics, appointment scheduling, digital medical record management, and health risk assessment.
React Node.js Python Machine Learning Tailwind CSS MongoDB
[Code Repository](#) [Live Demo](#)

FinTech & Analytics★ Featured Project

MarketLens: Unified Market Screener & Risk Analyzer
A unified financial market screening and risk analysis dashboard that evaluates stock metrics, algorithmic risk indicators, portfolio volatility, and real-time data visualization.
React.js Vite Python / FastAPI Financial APIs Chart.js PostgreSQL
[Code Repository](#) [Live Demo](#)

Let's Connect

Feel free to reach out for collaborations or project inquiries.

 jainamkamani95@gmail.com  [GitHub](#)  [LinkedIn](#)

© 2026 Jainam Kamani. All rights reserved.
Junagadh, Gujarat, India

 Jainam Kamani

Home Projects Contact

Get In Touch

Have a question or want to work together? Fill out the form below to send a direct message.

Your Name *

Krish Kaneria

Your Email *

krishkaneria07@gmail.com

Subject

Project Inquiry

Message *

Hello!

Send Message

Real-Time Live PreviewuseState Active

From Name:Krish Kaneria

Email Address:krishkaneria07@gmail.com

Subject:Project Inquiry

Message Content:

Hello!

Let's Connect

Feel free to reach out for collaborations or project inquiries.

 jainamkamani95@gmail.com  [GitHub](#)  [LinkedIn](#)

© 2026 Jainam Kamani. All rights reserved.
Junagadh, Gujarat, India

Original URL Of Web Browser: <http://localhost:5173/>
Input URL Of Web Browser: <http://localhost:5173/xyz>

 Jainam Kamani

Home Projects Contact

404 ERROR

 **Page Not Found**

Oops! The page path `/xyz` could not be found on this server.

It might have been moved, renamed, or perhaps it never existed.

[Return to Home](#) [Go Back](#) [View Projects](#)

Let's Connect

Feel free to reach out for collaborations or project inquiries.

 jainamkamani95@gmail.com  [GitHub](#)  [LinkedIn](#)

© 2026 Jainam Kamani. All rights reserved.
Junagadh, Gujarat, India

KEY QUESTIONS / ANALYSIS:

1) What is the role of each component in the overall UI structure?

In React, a component is a reusable piece of the user interface. Each component has a specific responsibility, making the application organized and easier to manage.

For example, in an e-commerce website:

- Header Component – Displays the logo, navigation menu, and search bar.
- Sidebar Component – Shows categories or filters.
- Product Card Component – Displays a product's image, name, price, and button.
- Footer Component – Contains contact information, links, and copyright.

The main App component combines all these smaller components to build the complete webpage.

2) How does React re-render when props change?

Props (properties) are values passed from a parent component to a child component.

When a parent component passes new props to a child:

1. The parent component updates.
2. React compares the new props with the previous props.
3. If the props have changed, React re-renders the child component.
4. React updates only the parts of the webpage that actually changed using the Virtual DOM, making updates efficient.

Example:

```
function Greeting(props) {  
  return <h1>Hello, {props.name}!</h1>;  
}
```

```
function App() {  
  return <Greeting name="Jainam" />;  
}
```

If name changes from "Jainam" to "Rahul", React re-renders only the Greeting component with the updated name.

3) Why is component reusability important in large-scale applications?

Component reusability is important because it improves development speed and makes applications easier to maintain.

Benefits:

- Reduces duplicate code – Write a component once and use it multiple times.
- Easier maintenance – Updating one reusable component updates it everywhere it is used.
- Consistency – The same design and behavior are maintained throughout the application.
- Faster development – Developers can build new pages by combining existing components.
- Better teamwork – Different developers can work on different components independently.

Example:

A Button component can be reused for:

- Login button
- Sign Up button
- Buy Now button
- Submit button

Instead of writing button code repeatedly, you create one reusable component and customize it using props.

Example:

```
function Button({ text }) {  
  return <button>{text}</button>; }  

```

// Reuse

```
<Button text="Login" />  
<Button text="Sign Up" />  
<Button text="Buy Now" />  

```

This makes React applications more scalable, easier to test, and simpler to maintain as they grow.

GITHUB REPOSITORY:

<https://github.com/9081495207/ITUE301-Advanced-Web-Development-Frameworks>